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250кит сдачи первого задания

1-3 недели II урока

прислать 0 и II

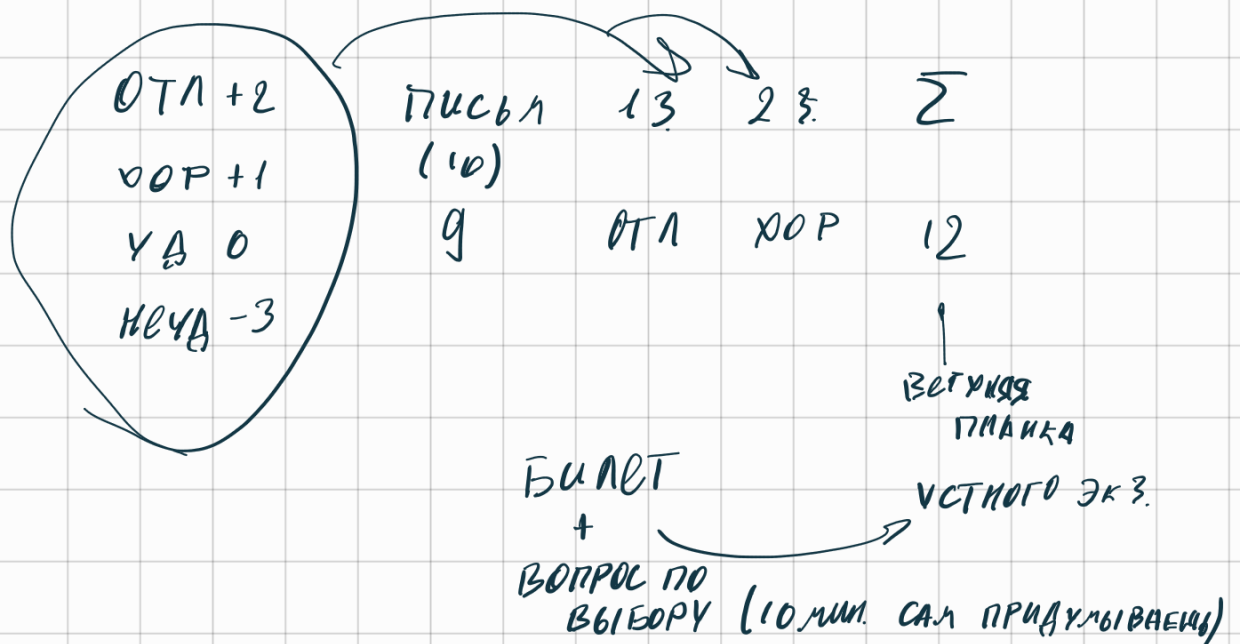
на к.р. печатать

Рисунк

Дано ; найти

ответ (формула + чис. отв.)

Физ. законы



302 ПК

Таблица производных	Таблица интегралов
1. $(C)' = 0$, $(x)' = 1$ $(x^2)' = 2x$	1. $\int dx = x + C$
2. $(x^a)' = ax^{a-1}$	2. $\int x^k dx = \frac{x^{k+1}}{k+1} + C$, $(k \neq -1)$
3. $(\sqrt{x})' = \frac{1}{2\sqrt{x}}$, $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$	3. $\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + C$, $\int \frac{dx}{x} = \ln x + C$
4. $(a^x)' = a^x \ln a$, $(e^x)' = e^x$	4. $\int a^x dx = \frac{a^x}{\ln a} + C$, $\int e^x dx = e^x + C$
5. $(\log_a x)' = \frac{1}{x \ln a}$, $(\ln x)' = \frac{1}{x}$	5. $\int \frac{dx}{\sin x} = \ln \left \operatorname{tg} \frac{x}{2} \right + C$,
6. $(\sin x)' = \cos x$	6. $\int \frac{dx}{\cos x} = \ln \left \operatorname{tg} \left(\frac{x}{2} + \frac{\pi}{4} \right) \right + C$
7. $(\cos x)' = -\sin x$	7. $\int \sin x dx = -\cos x + C$
8. $(\operatorname{tg} x)' = \frac{1}{\cos^2 x}$	8. $\int \cos x dx = \sin x + C$
9. $(\operatorname{ctg} x)' = -\frac{1}{\sin^2 x}$	9. $\int \frac{dx}{\cos^2 x} = \operatorname{tg} x + C$
10. $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$	10. $\int \frac{dx}{\sin^2 x} = -\operatorname{ctg} x + C$
11. $(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$	11. $\int \frac{dx}{\sqrt{a^2-x^2}} = \arcsin \frac{x}{a} + C$
12. $(\operatorname{arctg} x)' = \frac{1}{1+x^2}$	12. $\int \frac{dx}{\sqrt{x^2 \pm a^2}} = \ln \left x + \sqrt{x^2 \pm a^2} \right + C$
13. $(\operatorname{arcctg} x)' = -\frac{1}{1+x^2}$	13. $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left \frac{x-a}{x+a} \right + C$
	14. $\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \operatorname{arctg} \frac{x}{a} + C$

Правила дифференцирования и интегрирования		Некоторые тригонометрические формулы:	
а)	Правила дифференцирования	1	$\sin^2 x + \cos^2 x = 1$; $1 - \sin^2 x = \cos^2 x$; $1 - \cos^2 x = \sin^2 x$.
1	Производная суммы: $(u \pm v)' = u' \pm v'$	2	$\sin^2 x = \frac{1 - \cos 2x}{2}$, $\cos^2 x = \frac{1 + \cos 2x}{2}$,
2	Производная произведения: $(u \cdot v)' = u' \cdot v + u \cdot v'$	3	$\sin \alpha \cdot \sin \beta = \frac{1}{2} (\cos(\alpha - \beta) - \cos(\alpha + \beta))$

$$\vec{r}(t)$$

$$\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$$

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \vec{r}'_t = \dot{\vec{r}} = \frac{d\vec{r}}{dt}$$

$$\vec{a} = \dot{\vec{v}} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \frac{d\vec{r}}{dt} = \frac{d^2\vec{r}}{dt^2} = \ddot{\vec{r}} \quad \text{или } \ddot{\vec{r}}$$

$$x(t) = x_0 + v_0 t + \frac{at^2}{2} \text{ полином. формул.}$$

$$a = \frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx}$$

$$a dx = v dv$$

$$\int a dx = \int v dv$$

$$ax + C_1 = \frac{v^2}{2} + C_2$$

$$ax = \frac{v^2}{2} + C$$

$$ax_0 = \frac{v_0^2}{2} + C$$

$$C = ax_0 - \frac{v_0^2}{2}$$

$$ax = \frac{v^2}{2} + ax_0 - \frac{v_0^2}{2}$$

$$a(x - x_0) = \frac{v^2 - v_0^2}{2}$$

$$a dx = v dv$$

$$\int_{x_0}^x a dx = \int_{v_0}^v v dv$$

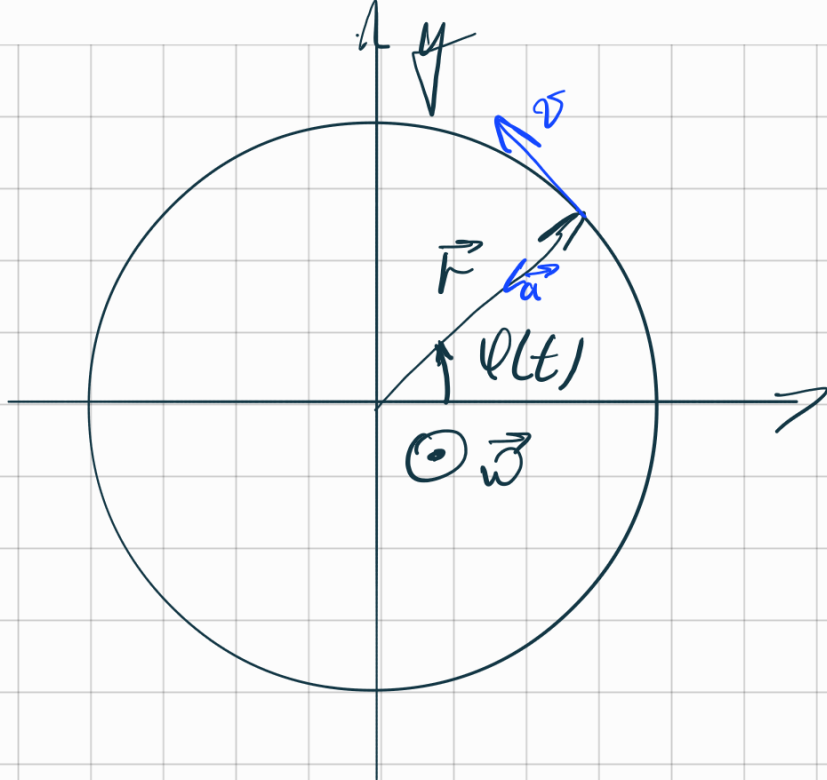
$$ax \Big|_{x_0}^x = \frac{v^2}{2} \Big|_{v_0}^v$$

$$a(x - x_0) = \frac{v^2 - v_0^2}{2}$$

$$x(t) = v_0 \cos \alpha t$$

$$y(t) = v_0 \sin \alpha t - \frac{gt^2}{2}$$

$$y = \operatorname{tg} \alpha \cdot x - \frac{g x^2}{2 v_0^2 \cos^2 \alpha}$$



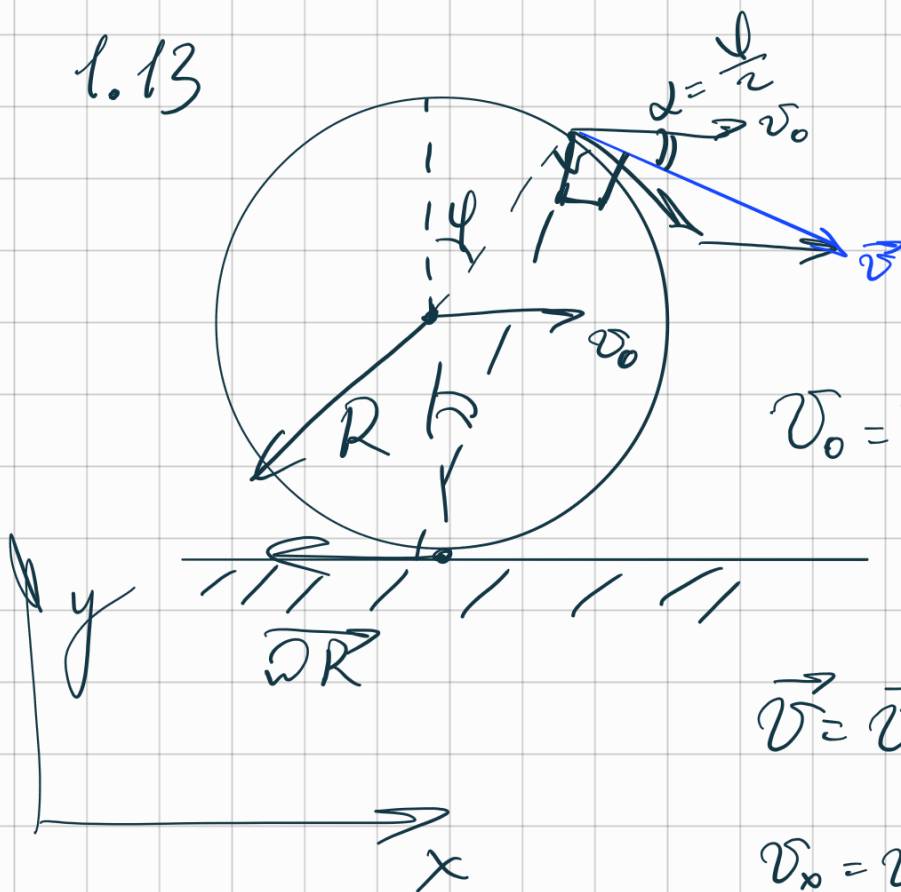
$$\dot{\omega} = \dot{\dot{\varphi}} - \text{yml. step.}$$

$$\mathcal{E} = \dot{\omega} = \dot{\dot{\varphi}} - \text{yml. yml.}$$

$$\times \vec{v} = [\vec{\omega} \times \vec{r}]$$

$$\vec{a} = [\vec{\omega} \times \vec{v}]$$

1.13

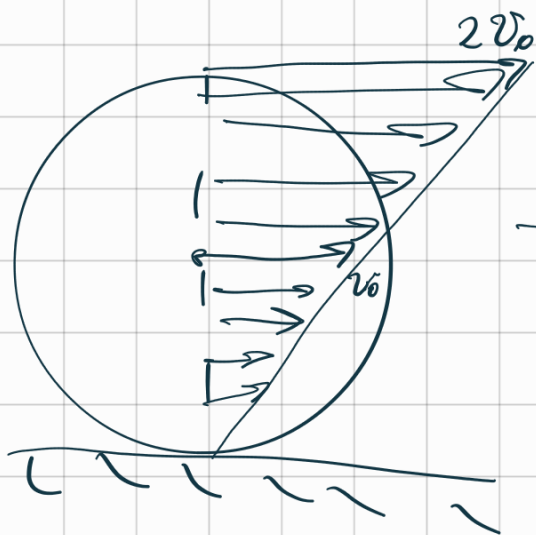


$$v_0 = \omega R \text{ - See previous.}$$

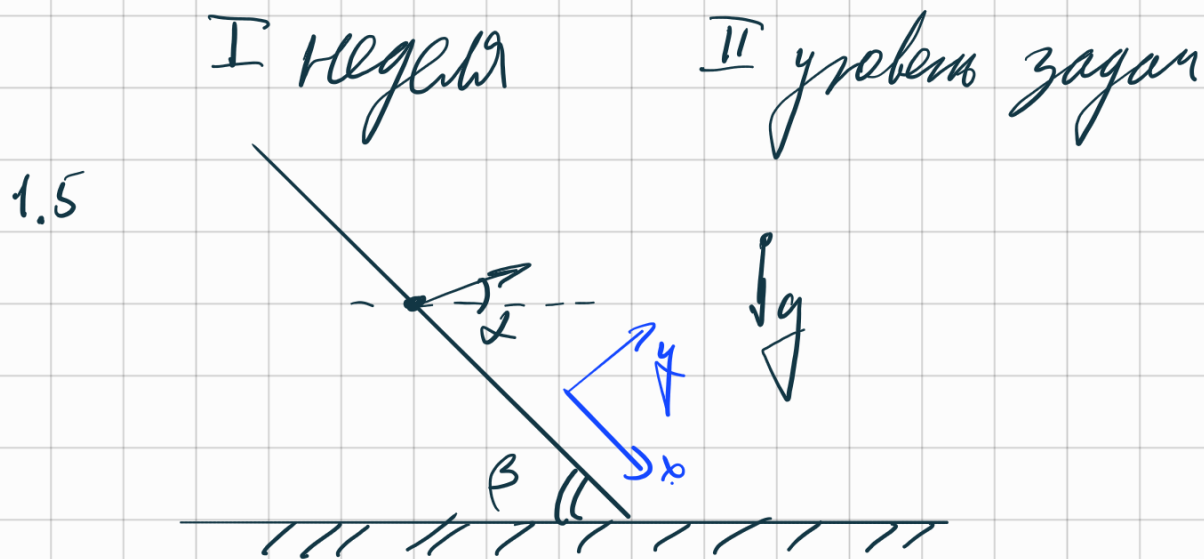
$$\vec{v} = \vec{v}_0 + [\vec{\omega} \times \vec{R}]$$

$$v_x = v_0 + v_0 \cos \psi$$

$$v_y = 0 + v_0 \sin \psi$$



$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{v_0^2(1 + \cos \psi)^2 + v_0^2 \sin^2 \psi} = v_0 \sqrt{2(1 + \cos \psi)}$$



Дано:

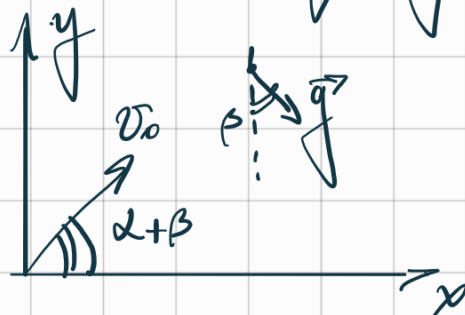
$$\beta = 45^\circ$$

Найти: α ; $x(\max)$

Решение:

введем С.О., где

x паралл. склону, а y перп. ему:



тогда в этих координатах

$$x = v_0 t \cos(\alpha + \beta) + \frac{g t^2}{2} \cdot \sin \beta$$

$$v_x = v_0 \cos(\alpha + \beta) + g t \cdot \sin \beta$$

$$y = v_0 t \sin(\alpha + \beta) - \frac{g t^2}{2} \cos \beta$$

$$v_y = v_0 \sin(\alpha + \beta) - g t \cdot \cos \beta$$

$$t = \frac{2 v_0 \sin(\alpha + \beta)}{g \cos \beta} \quad \text{— время полета}$$

$$x = v_0 \cos(\alpha + \beta) \cdot \frac{2 v_0 \sin(\alpha + \beta)}{g \cos \beta} + \frac{g \sin \beta}{2} \cdot \frac{4 v_0^2 \sin^2(\alpha + \beta)}{g^2 \cos^2 \beta}$$

$$\frac{x \cdot g \cdot \cos \beta}{v_0^2} = 2 \cos(\alpha + \beta) \cdot \sin(\alpha + \beta) + 2 \operatorname{tg} \beta \sin^2(\alpha + \beta) \quad (\beta = 45^\circ \Rightarrow \operatorname{tg} \beta = 1)$$

$$\frac{x g \cos \beta}{v_0^2} = \sin(2(\alpha + \beta)) + 2 \sin^2(\alpha + \beta) \quad \left| \cdot \frac{1}{2} \right. \leftarrow \text{узнаем новую}$$

$$0 = 2 \cos(2(\alpha + \beta)) + 4 \cos(\alpha + \beta) \sin(\alpha + \beta)$$

экстремума
функции угла $x(\alpha)$

$$0 = \cos(2(\alpha + \beta)) + \sin(2(\alpha + \beta)) \Leftrightarrow \operatorname{tg}(2(\alpha + \beta)) = -1$$

$$2(\alpha + \beta) = \frac{3\pi}{4}$$

или

$$2(\alpha + \beta) = -\frac{\pi}{4}$$

$$\alpha + \beta = \frac{3\pi}{8}$$

$$\beta = \frac{\pi}{4}$$

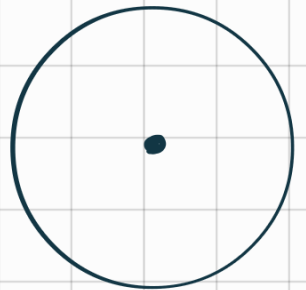
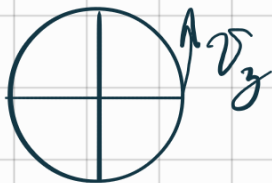
$$\alpha + \beta = -\frac{\pi}{8}$$

норм. корни
т.к. корни
бывают как
сложными

$$\alpha = \frac{\pi}{8}$$

$$\text{Ответ: } \frac{\pi}{8}$$

1.11



Дано:

$$R_n = 3,8 \cdot 10^5 \text{ км}$$

$$R_z = 6,38 \cdot 10^3 \text{ км}$$

$$T_z = 86164 \text{ с}$$

$$T_n = 2360591 \text{ с}$$

Найти: v_T

Решение:

Скорость тела v_T равна $v_T = v_n - v_z$,
где v_n - скорость углов по орбите, v_z - скорость
вращения поверхности Земли вокруг своей оси.

$$v_n = \frac{2\pi R_n}{T_n}$$

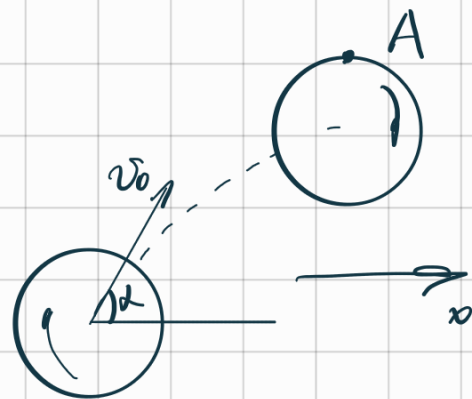
$$v_z = \frac{2\pi R_z}{T_z}$$

$$v_T = 2\pi \left(\frac{R_n}{T_n} - \frac{R_z}{T_z} \right) = 0,55 \frac{\text{км}}{\text{с}}$$

$$\text{Ответ: } 0,55 \frac{\text{км}}{\text{с}}$$

1.21

Решение:



Дано:

 $R; \omega; \alpha; v_0$ Найти: ρ

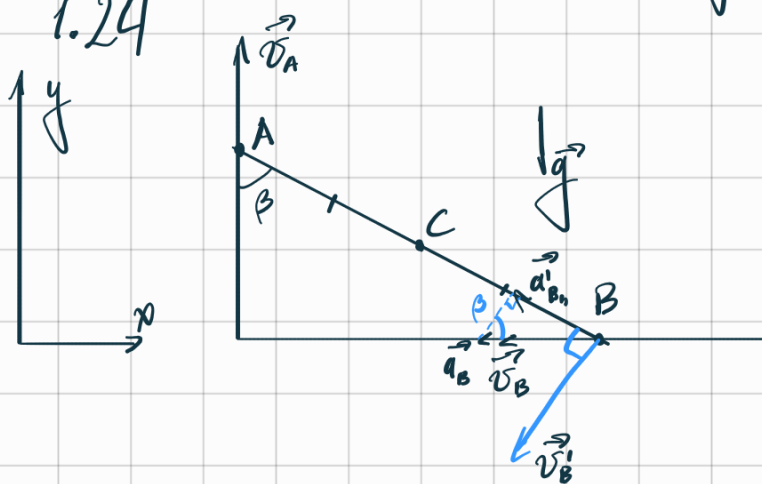
Т.к. диск находится в крайней точке, скорость его центра масс равна $v_0 \cos \alpha$, т.к. диск крутится по часовой стрелке и A находится в наивысшем положении (относительно ц.м.), то ее скорость относительно ц.м. (спроецированная на горизонт. ось Ox) равна ωR , а относительно земли $v_A = v_0 \cos \alpha + \omega R$ на A действует g и ц.с. ускорение $\omega^2 R$, тогда

$$a_A = g + \omega^2 R \left(\rho = \frac{v_A^2}{a_A} \right) \Rightarrow \rho = \frac{(v_0 \cos \alpha + \omega R)^2}{g + \omega^2 R}$$

(ц.с. ускорение сонаправлено с g)

Ответ: $\frac{(v_0 \cos \alpha + \omega R)^2}{g + \omega^2 R}$

1.24



Решение:

Т.к. $v_A = \text{const}$ введем ИСО, в которой A неподвижна

$$\vec{v}_B = \vec{v}_A + \vec{v}_B' \quad (\vec{v}_B' \perp l)$$

$$\vec{a}_B = \vec{a}_B'$$

спроецируем $\vec{v}_B$ на Oy :

$$v_A - v_B' \cdot \sin \beta = 0 \quad v_B' = \frac{v_A}{\sin \beta}$$

(стрелка вращается вокруг A)

$$a_{B_n}' = \frac{v_B'^2}{l} \quad a_{B_n}' = a_B' \cdot \sin \beta$$

$$a_B = a_B' = \frac{a_{B_n}'}{\sin \beta} = \frac{1}{\sin \beta} \cdot \frac{1}{l} \cdot \frac{v_A^2}{\sin^2 \beta} = \frac{v_A^2}{l \sin^3 \beta}$$

Дано:

 $l; v_A = \text{const}; \beta$ Найти: $\vec{a}_B$

Т.к. ступень "вращается" вокруг $\{A\}$, скорости и ускорения в любой его точке сонаправлены, тогда $a_C = \frac{1}{2} a_B = \frac{v_A^2}{2l \sin^3 \beta}$

$$\text{Ответ: } \frac{v_A^2}{2l \sin^3 \beta}$$

II очередь

0 уровень задач

2.1

Дано:

$$\dot{x} = \frac{x}{\tau}$$

$$\tau = 10 \text{ с.}$$

$$x_0 = 1 \text{ м.}$$

Найти: $x(t=\tau)$, $a(t=\tau)$

Решение:

$$1. \quad v = \frac{dx}{dt} = \frac{x}{\tau}$$

$$\frac{1}{x} dx = \frac{1}{\tau} dt$$

$$\ln(x) = \frac{t}{\tau} + C$$

$$\ln(x_0) = C$$

$$x_0 = 1 \Rightarrow C = 0$$

$$x = e^{\frac{t}{\tau}} \quad x(t=\tau) = e$$

$$2. \quad v = \frac{dx}{dt} = \frac{de^{\frac{t}{\tau}}}{dt}$$

$$v = \frac{1}{\tau} e^{\frac{t}{\tau}}$$

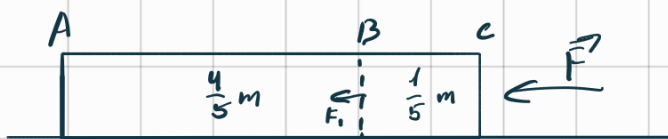
$$a = \frac{dv}{dt} = \frac{1}{\tau} \frac{de^{\frac{t}{\tau}}}{dt}$$

$$a = \frac{1}{\tau^2} \cdot e^{\frac{t}{\tau}}$$

$$a(t=\tau) = \frac{e}{\tau^2} = \frac{e}{100}$$

$$\text{Ответ: } e \text{ м.}; \frac{e}{100} \frac{\text{м}}{\text{с}^2}$$

2.1



Дано:

$$F; AB = \frac{4}{5} l$$

$$m; l$$

Найти: F_1

Решение:

$$a_{AC} = a_{BC} = a_{AB}$$

$$\vec{F}_{AB} = -\vec{F}_1 \quad F_{AB} = m_{AB} \cdot a_{AB}$$

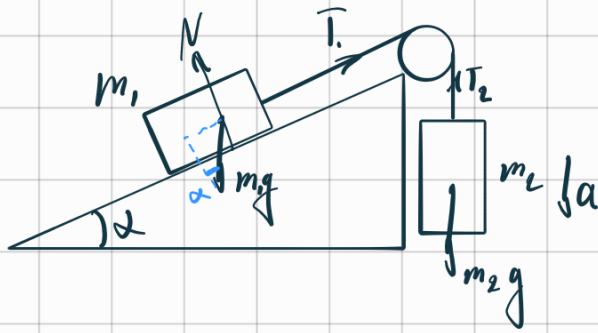
$$AB = \frac{4}{5} l \Rightarrow m_{AB} = \frac{4}{5} m$$

$$F_{AB} = \frac{4}{5} m a_{AC} = \frac{4}{5} F$$

$$F_1 = -\frac{4}{5} F$$

$$\text{Ответ: } -\frac{4}{5} F$$

2.5



Решо:

 m_1, m_2, α Иском: T, a (считаем трения нет,
т.е. идеальное тело)

Решение:

$$T_1 = T_2 = T$$

Из уравнений, что m_2
опускается:

$$m_2 a = m_2 g - T$$

$$m_1 a = T - m_1 g \cdot \sin \alpha$$

$$a(m_2 + m_1) = g(m_2 - m_1 \sin \alpha)$$

$$a = g \frac{m_2 - m_1 \sin \alpha}{m_1 + m_2}$$

$$T = m_2 g - m_2 g \frac{m_2 - m_1 \sin \alpha}{m_1 + m_2} = m_2 g \left(\frac{m_1 + m_2 - m_2 + m_1 \sin \alpha}{m_1 + m_2} \right) = \frac{m_1 m_2 g}{m_1 + m_2} (1 + \sin \alpha)$$

$$\text{Ответ: } g \frac{m_2 - m_1 \sin \alpha}{m_1 + m_2}; \frac{m_1 m_2 g}{m_1 + m_2} (1 + \sin \alpha)$$

$$1.17 \quad \int_0^T |v| dt = \int v_0 \sqrt{2(1+\cos\varphi)} dt =$$

$$= v_0 \sqrt{2 \frac{1+\cos\varphi}{2}} dt =$$

$$\varphi(t) = \omega t - \pi$$

$$d\varphi = \omega dt$$

$$= 2v_0 \int_0^T \cos \frac{\varphi}{2} dt =$$

$$\cos \frac{\varphi}{2} = \sqrt{\frac{1+\cos\varphi}{2}}$$

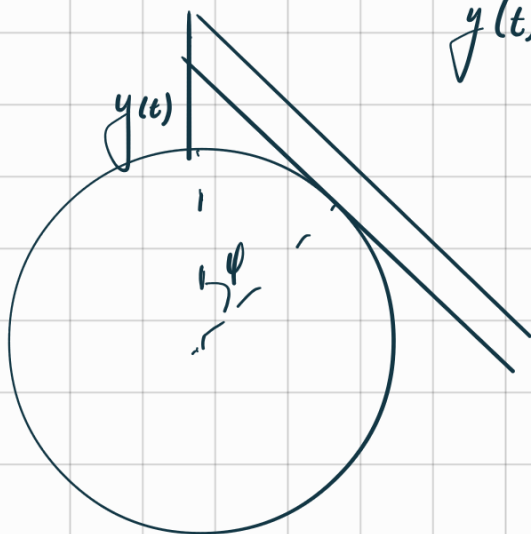
$$= 2v_0 \int_0^T \cos \frac{\omega t - \pi}{2} dt =$$

$$= 2v_0 \sin \frac{\omega t - \pi}{2} \cdot \frac{2}{\omega} \Big|_0^T =$$

$$= 4 \frac{v_0}{\omega} \left(\sin \frac{\omega \frac{2\pi R}{v_0} - \pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right) =$$

$$= 4R \left(\sin \frac{\pi}{2} - \sin \left(-\frac{\pi}{2} \right) \right) = 8R$$

1.12



$y(t) = ?$

$$\dot{\varphi} = \frac{2\pi}{T_y}$$

$$\cos\varphi = \frac{R}{R+y}$$

$$y = R \frac{1 - \cos\omega t}{\cos\varphi \omega t}$$

$$y \approx R(1 - \cos\omega t)$$

$$\dot{y} = R \sin\omega t \cdot \omega \quad \ddot{y} = \omega^2 R \cos\omega t = 3,4 \frac{\text{cm}}{\text{s}^2}$$

$$H = \frac{\dot{y} \tau}{2}$$

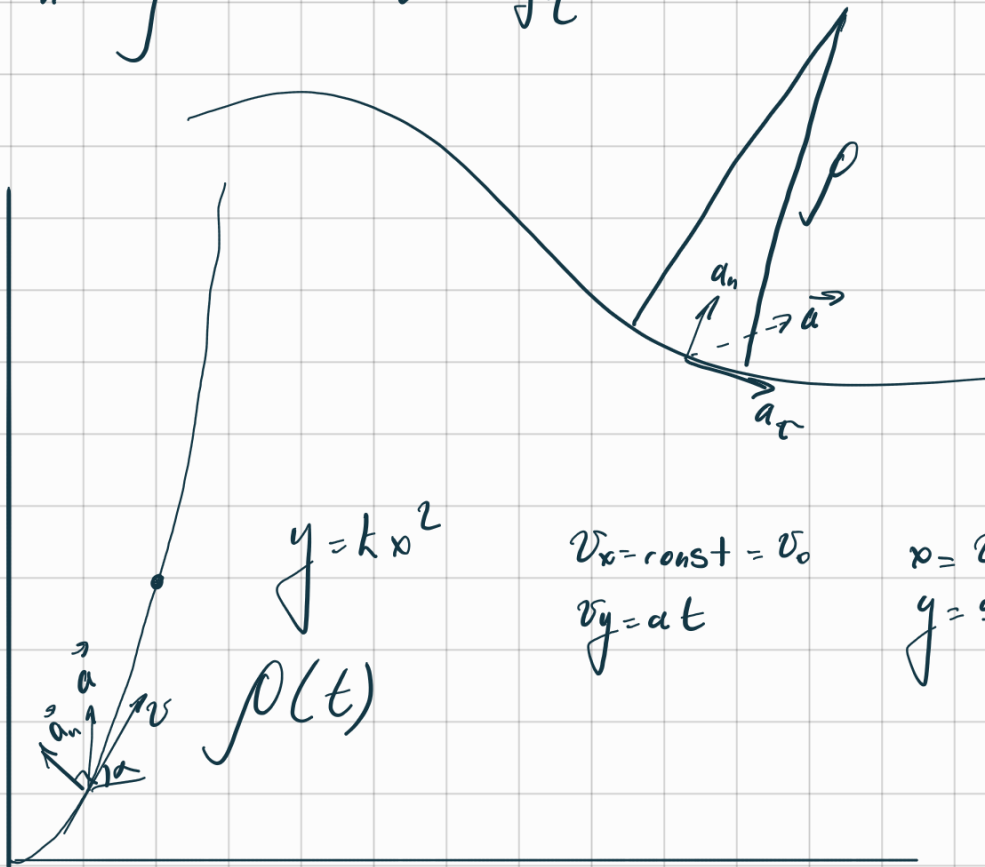
$$T = \sqrt{\frac{2H}{\omega^2 R}} \approx 5 \text{ mm}$$

$$\cos \omega t = 1 - \frac{(\omega t)^2}{2}$$

$$y = R \frac{\frac{(\omega t)^2}{2}}{1 - \frac{(\omega t)^2}{2}} = R \frac{(\omega t)^2}{2} \cdot \frac{2}{2 - (\omega t)^2} = R \frac{(\omega t)^2}{2 - (\omega t)^2}$$

$$a_n = \frac{v^2}{\rho}$$

$$a_\tau = \frac{dv}{dt}$$



$$v_x = \text{const} = v_0$$

$$v_y = at$$

$$x = v_0 t$$

$$y = \frac{at^2}{2} = kx^2 = v_0^2 t^2 k$$

$$v_0 = \sqrt{\frac{a}{2k}}$$

$$v^2 = v_x^2 + v_y^2 = \frac{a}{2k} + a^2 t^2$$

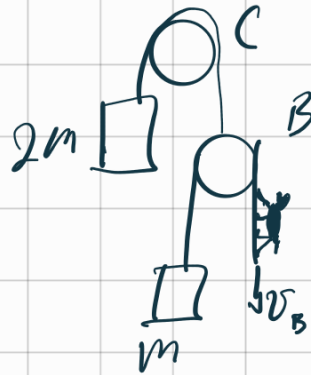
$$a_n = a \cos \alpha$$

$$\cos \alpha = \frac{v_x}{v}$$

$$\vec{p} = m \vec{v}$$

$$m \vec{a} = m \frac{d\vec{v}}{dt} = \frac{d\vec{p}}{dt} = \vec{F}$$

2.17



Условия:

$$T_1 - mg = ma_1$$

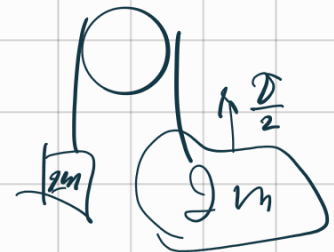
$$T_2 - mg = ma_2$$

$$T_1 = T_2 = T$$

(нить нерастяжима, невесомая)

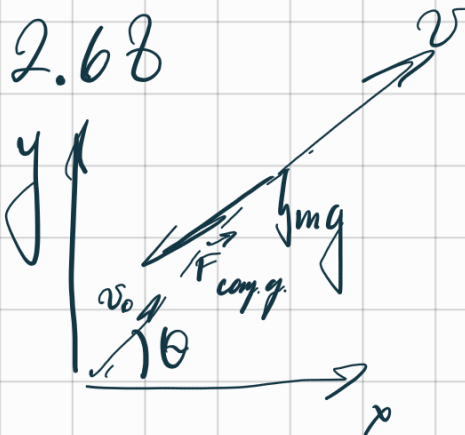
$$v_{\text{отн B}} = \frac{v_B}{2}$$

$$v_B - v_{\text{отн B}} = v_{\text{отн B}}$$



$$v_{2m} = \frac{v}{4}$$

2.68



$$t(K_{\max})$$

$$L(K_{\max})$$

$$m \frac{dv_x}{dt} = -\beta v_x$$

$$m \frac{dv_y}{dt} = -\beta v_y - mg$$

$$v_x(t) = v_0 \cos \theta e^{-\frac{\beta t}{m}}$$

$$x(t) = \int_0^t v_0 \cos \theta e^{-\frac{\beta t}{m}} dt =$$

$$= \frac{m v_0}{\beta} \cos \theta (1 - e^{-\frac{\beta t}{m}})$$

$$m \frac{dv_y}{dt} = -\beta (v_y + \frac{mg}{\beta})$$

$$\frac{dv_y + \frac{mg}{\beta}}{v_y + \frac{mg}{\beta}} = -\frac{\beta}{m} dt$$

$$\ln(v_y + \frac{mg}{\beta}) = -\frac{\beta}{m} t$$

$$\ln v_y + \frac{mg}{\beta}$$

2.43



$$F_{TP} = \mu N$$

$$\Pi_3.11$$

$$ma_n = N = m \frac{v^2}{R}$$

$$ma_\tau = -F_{TP} = -\mu \cdot N = -\mu m \frac{v^2}{R}$$

$$\frac{dv}{dt} = \frac{dv}{dx} \frac{dx}{dt} = v \frac{dv}{dx} = -\mu \frac{v^2}{R}$$

$$\int \frac{dv}{v} = -\int \frac{\mu}{R} dx$$

$$\ln \frac{v}{v_0} = -\frac{\mu l}{R}$$

$$v = v_0 e^{-\frac{\mu l}{R}}$$

$$v_k = v_0 e^{-\pi \mu}$$

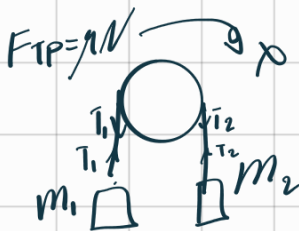
$$\frac{dv}{dt} = -\mu \frac{v^2}{R}$$

$$\int_{v_0}^v \frac{dv}{v^2} = \int \frac{-\mu}{R} dt$$

$$\frac{1}{v_0} - \frac{1}{v} = -\frac{\mu t}{R}$$

$$v(t) = \frac{v_0 R}{R + v_0 \mu t}$$

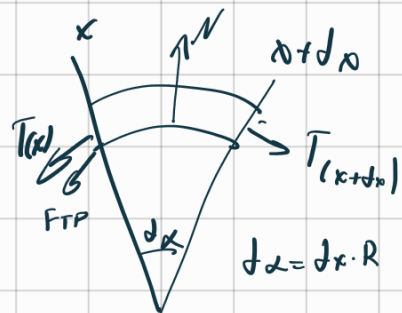
2.57



б. габарити

$$T_1 = m_1 g$$

$$T_2 = m_2 g$$



КАСАТЕЛЬНАЯ

$$T(x+dx) \cos \frac{d\alpha}{2} = T(x) \cos \frac{d\alpha}{2} + F_{TP}$$

$$dT = T(x+dx) - T(x) = \mu N$$

НОРМАЛЬ

$$N = T_x \sin \frac{d\alpha}{2} + T_{(x+dx)} \sin \frac{d\alpha}{2} = T d\alpha$$

$$dT = \mu T d\alpha$$

$$\int_{T_1}^{T_2} \frac{dT}{T} = \int_0^\pi \mu d\alpha$$



$$\frac{T_2}{T_1} = e^{\mu\pi} = \frac{m_2(g-a)}{m_1(g+a)}$$

в равновесии

$$e^{-\pi\mu} \leq \frac{m_2}{m_1} \leq e^{\pi\mu}$$

$$\vec{F} = \frac{d\vec{P}}{dt} = \frac{d(m\vec{v})}{dt}$$

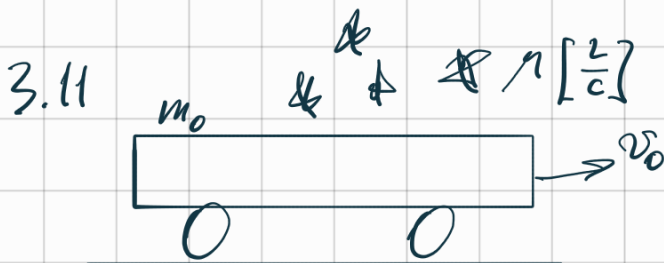
$$m \frac{d\vec{v}}{dt} = \vec{F} + \vec{u} \frac{dm}{dt}$$

у-е релятивистского



$$m = m_0 e^{-\frac{v}{u}} - \text{ф-ла Циолковского}$$

$$v = u \ln \frac{m_0}{m}$$



1) через нарастающую

$$m(t) = m_0 + \mu t$$

$$m_0 v_0 = (m_0 + \mu t) u$$

$$u = \frac{m_0 v_0}{m_0 + \mu t} \xrightarrow{t \rightarrow \infty} 0$$

$$x(t) = \frac{m_0 v_0}{\mu} \int \frac{dt}{\frac{m_0}{\mu} + t} = \frac{m_0 v_0}{\mu} \ln \left(1 + \frac{\mu t}{m_0} \right) \xrightarrow{t \rightarrow \infty} \frac{m_0 v_0}{\mu}$$

2) через сохраняющуюся

$$\int P = F dt = 0$$

$$(m_0 + \mu t)(v + dv) = m_0 v$$

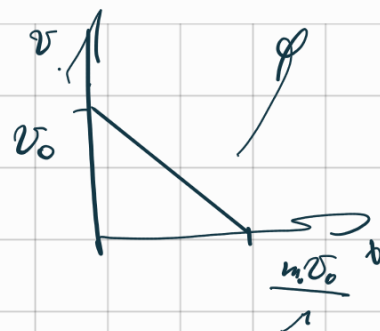
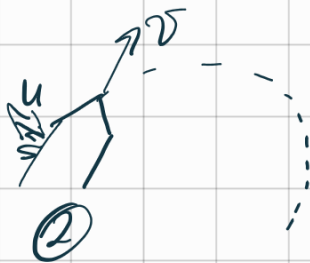
$$m_0 v + m_0 dv + \mu v dt + \mu t dv = m_0 v$$

$$m_0 dv = -\mu v dt = -\mu dx \quad m_0 v = -\mu x$$

$$\frac{dv}{v} = -\frac{\mu}{m_0} dt$$

$$v_{\text{lim}} = v_0 e^{-\frac{\mu t}{m_0}} \xrightarrow{t \rightarrow \infty} 0$$

3.43



$$1) \underbrace{(m+dm)(v+dv)}_{P_{\text{ПРЯК}}} + \underbrace{(-dm)(v+u)}_{P_{\text{ТОЛЧБА}}} - m v = 0$$

$$m dv - dm u = 0$$

$$\ln \frac{m}{m_0} = \frac{1}{u} (v - v_0)$$

$$m_1 = m_0 e^{-\frac{2v_0}{u}}$$

$$2) (m+dm)(\vec{v}+d\vec{v}) + (-dm)(\vec{v}+u\vec{u}) = m\vec{v}$$

$$m d\vec{v} - dm u\vec{u} = 0$$



$$-m dv - u dm = 0$$

$$dx = \frac{dv}{v_0}$$

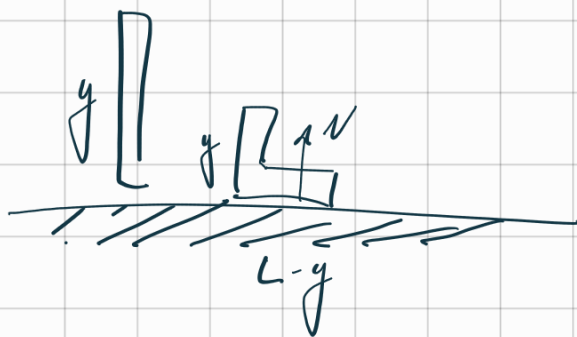
$$\bullet -m v_0 dx = u dm$$

$$\int_{m_0}^m \frac{dm}{m} = - \int_{v_0}^v \frac{dv}{v_0}$$

$$\ln \frac{m}{m_0} = -\frac{u}{v_0} v$$

$$m = m_0 e^{-\frac{u}{v_0} v}$$

4.55



$$v = m \frac{L-y}{L} g + \frac{dP_{y1}}{dt} = m \frac{L-y}{L} g + \frac{dm \cdot v}{dt} = m \frac{L-y}{L} g + \frac{m}{L} \frac{dy}{dt} \cdot \frac{2g(L-y)}{v_y} = m \frac{L-y}{L} g + \frac{m}{L} \cdot 2g(L-y) \cdot \frac{1}{v_y}$$

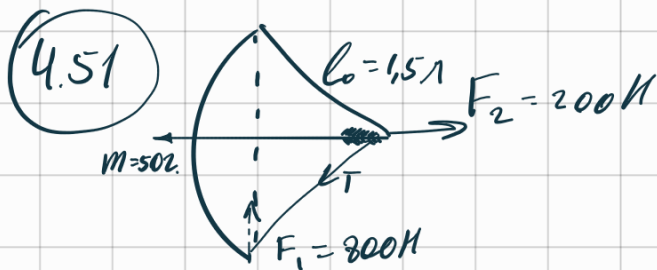
$$= 3 \frac{m}{L} (L-y) g$$

$$\delta A = (\vec{F} \cdot d\vec{r})$$

$$A = \int \vec{F} \cdot d\vec{r}$$

$$\delta a = m \frac{d\vec{v}}{dt} \cdot d\vec{r} = m \vec{v} \cdot d\vec{v} = \int \frac{m v^2}{2} = \int \left(\frac{p^2}{2m} \right) = \int K$$

$$\Delta K = A_{\text{внешн}}$$



Т.е. и из А нормальная
деформация при растяжении

$$F = \text{const} = T = F_1$$

$$F(x) = 2T \sin \varphi = 4 F_1 \frac{x}{L_0} \quad x_{\text{max}} = \frac{F_2 L_0}{4 F_1}$$

$$A = \int_0^{x_{\text{max}}} F_x dx = \frac{F_2^2 L_0}{8 F_1} = \frac{m v_0^2}{2}$$

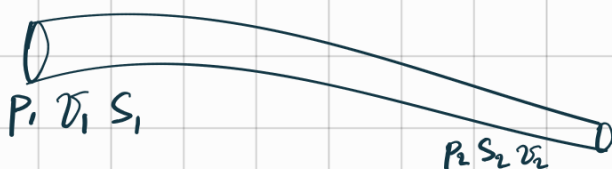
$$L_{\text{max}} = \frac{v_0^2}{g} = \frac{F_2^2 L_0}{4 F_1 m g} = \frac{100^2}{4 \cdot 200 \cdot 0,05 \cdot 10} = \frac{100 \cdot 45}{40} = 25 \cdot 4,5 = 37,5$$

$$\Delta K = A_{\text{внешн}} = A_{\text{конс}} + A_{\text{внешн}} = -\Delta U + A_{\text{внешн}}$$

$$\exists \text{ концеп. } F \quad \delta U = -\delta A$$

$$\Delta E = \Delta(K+U) = A_{\text{внешн}}$$

и снова гидродинамика...



математический подход

$$\frac{dm_1}{dt} = -\frac{dm_2}{dt}$$

$$\frac{m}{2} (v_2^2 - v_1^2) + (u_2 - u_1) = P_1 S_1 dx_1 - P_2 S_2 dx_2$$

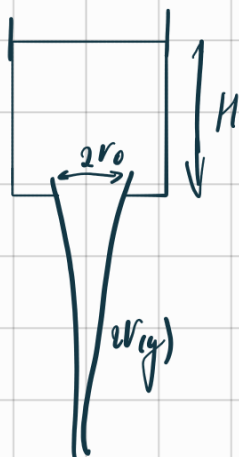
$$\frac{v_2^2}{2} - \frac{v_1^2}{2} + \frac{u_2 - u_1}{m} = \frac{P_1}{\rho_1} - \frac{P_2}{\rho_2}$$

$$\frac{P_1}{\rho_1} + \frac{u_1}{m} + \frac{v_1^2}{2} = \frac{P_2}{\rho_2} + \frac{u_2}{m} + \frac{v_2^2}{2} = \text{const}$$

$$\frac{\rho_1 S_1 dx_1}{dt} = \frac{\rho_2 S_2 dx_2}{dt}$$

$$\rho_1 S_1 v_1 = \rho_2 S_2 v_2$$

14.11



$$P_0 - 0 + \rho \frac{v^2}{2} = P_0 - \rho g y + \frac{\rho v_y^2}{2} = P_0 + \rho g H$$

$$\pi r_0^2 v_0 = \pi r_y^2 v_y$$

$$v_y = \sqrt{2g(H+y)}$$

$$v_y = r_0 \sqrt{\frac{v_0^2}{r_y^2}} = r_0 \sqrt{\frac{H}{H+y}}$$

BCL 🤪

Угловы мом

$$\vec{r}_c = \frac{\sum m_i \vec{r}_i}{\sum m_i} \rightarrow \frac{\int \vec{r} dm}{M}$$

$$\vec{v}_c = \frac{\sum m_i \vec{v}_i}{\sum m_i} = \frac{\vec{P}}{M}$$

$K_{\text{обм}} = K_c + K_{\text{отн}}$ - Теорема Кёнига

1) $K_{\text{отн}} \rightarrow 0$ для нел. углов. углов

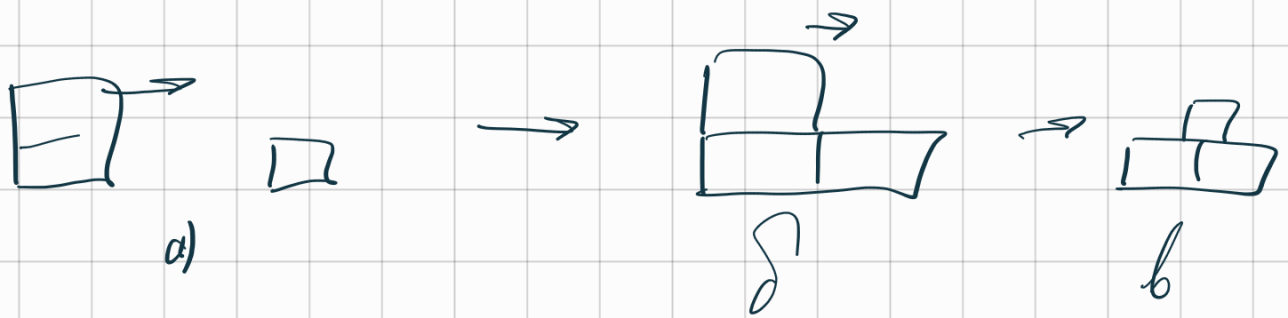
2) $K_{\text{отн}} = \text{const}$ для углов. углов

4.25 $\begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix}$ -- уг. 12
-- уг. 11

$$t = \sqrt{\frac{2L}{g}} = \sqrt{\frac{L}{g}}$$

$t' = \sqrt{2} t$ - для нел. углов. углов

4.47



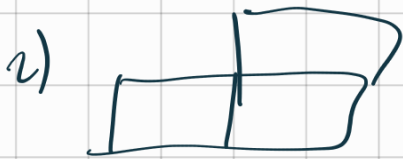
$$P_a = P_b$$

$$m v_0 + m v_0 = 2m v + m v_0$$

$$v = \frac{v_0}{2}$$

$$A_{TP} = \int_0^l F_{TP} dx = \int_0^l k N(x) dx =$$

$$= \int_0^l k mg \frac{x}{2} dx = k mg \frac{l}{2}$$



$$P_2 = P_1$$

$$2m v + m v_0 = 3m v_k$$

$$v_k = \frac{2}{3} v_0$$

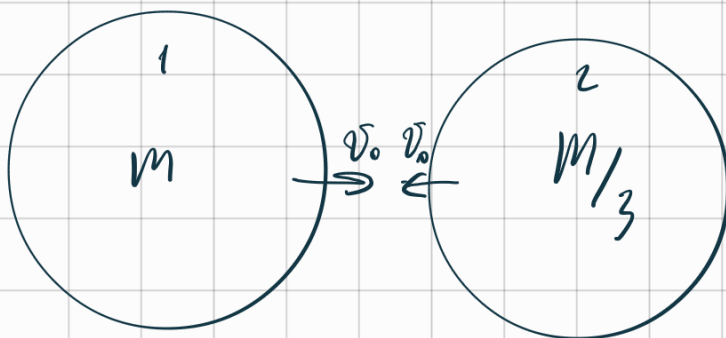
$$A_{TP} = -\Delta K_{TB} = -\left(\frac{3}{2} m \cdot \left(\frac{2}{3} v_0 \right)^2 - \frac{2}{2} m \left(\frac{v_0^2}{2} \right) - m \frac{v_0^2}{2} \right) = \frac{m v_0^2}{12}$$

$$l = \frac{v^2}{6kg}$$

Еще про Т. Кеминга

$$K = K_c + K_{отн} = \frac{P^2}{2A} + K_{отн}$$

4.80



$$F_{упр} = k x^{\frac{3}{2}}$$

$$x_{max} = ?$$

$$K = \frac{P^2}{2A} + K_{отн}$$

$$K_{отн} = \frac{m v_k^2}{2} + \frac{m}{3} \frac{v_{rel}^2}{2}$$

П (только эта составляющая
на деформацию)

$$v_c = \frac{mv_0 - \frac{m}{3}v_0}{m + \frac{m}{3}} = \frac{v_0}{2}$$

$$v_{1c} = v_0 - v_c = \frac{v_0}{2}$$

$$v_{2c} = -v_0 - v_c = -\frac{3v_0}{2}$$

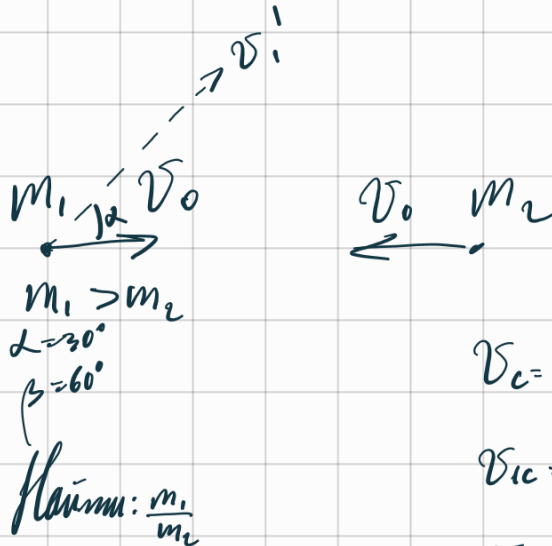
$$K_{отн} = \frac{m}{2} \frac{v_0^2}{4} + \frac{m}{3} \frac{9v_0^2}{4} = m \frac{v_0^2}{2}$$

$$P_{max} = 2 \int_0^{x_{max}} k x^{\frac{3}{2}} dx = \frac{4k}{5} x_{max}^{\frac{5}{2}}$$

↑
т.к. грав. сила

$$x_{max} = \left(\frac{5 m v_0^2}{8 k} \right)^{\frac{2}{5}}$$

4.90

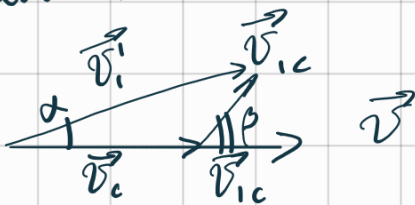


$$v_c = \frac{m_1 v - m_2 v}{m_1 + m_2} = v \frac{m_1 - m_2}{m_1 + m_2} = v \frac{x-1}{x+1}$$

$$v_{1c} = v - v_c = v \left(1 - \frac{x-1}{x+1} \right) = \frac{2v}{x+1}$$

$$v_{2c} = -v - v_c = v \left(1 - \frac{x-1}{x+1} \right) = \frac{-2vx}{x+1}$$

для m.



$$\vec{v}_1' = \vec{v}_c + \vec{v}_{1c}'$$

$$v_1' \cos \alpha = v_c + v_{1c}' \cos \beta$$

$$v_1' \sin \alpha = v_{1c}' \sin \beta$$

$$\operatorname{ctg} \alpha = \frac{v_c}{v_{1c}' \sin \beta} + \operatorname{ctg} \beta = \frac{x-1}{x+1} \frac{x+1}{2 \sin \beta} + \operatorname{ctg} \beta$$

$$x = 1 + 2 \sin \beta (\operatorname{ctg} \alpha - \operatorname{ctg} \beta) = 3$$

угол упругий $K_{отн} = K'_{отн}$

$$P_{отн} = 0 = m_1 v_{1c} + m_2 v_{2c}$$

$$P'_{отн} = m_1 v_{1c}' + m_2 v_{2c}'$$

$$\left. \begin{aligned} |v_{1c}'| &= |v_{1c}| \\ |v_{2c}'| &= |v_{2c}| \end{aligned} \right\} \Rightarrow$$

Пример

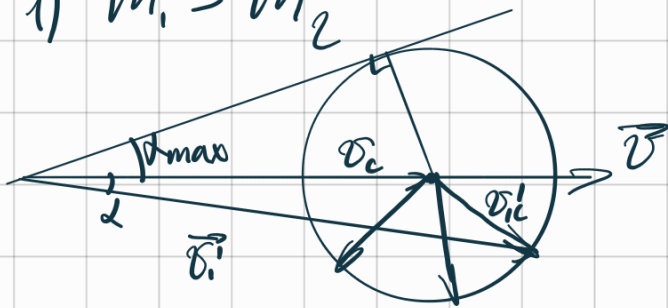


$$m_2 \quad v_c = \frac{m_1 v}{m_1 + m_2}$$

$$v_1 = v \left(1 - \frac{m_1}{m_1 + m_2} \right) = \frac{v m_2}{m_1 + m_2}$$

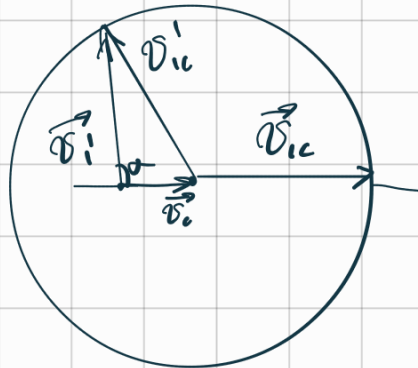
$$v_2 = -v_c$$

$$1) m_1 > m_2$$



$$\sin \alpha_{\max} = \frac{v'_{ic}}{v_c} = \frac{m_2}{m_1}$$

$$2) m_1 < m_2$$



4.142



Li

$$x = \frac{m_{Li}}{m_{Be}} = \frac{7}{9} \quad K = 18 \Rightarrow B = 1.6 \cdot 10^{-19} \cdot 18 \text{ Дж} \quad \psi_{\max} = 15^\circ$$

$$v_c = \frac{v}{1+x}$$

$$v_{ic} = v - v_c = v \left(1 - \frac{1}{1+x}\right) = \frac{v x}{1+x}$$

$$v_{ic} = -v_c$$

$$K_{\text{отн}} = \frac{m_{Be} v_{ic}^2}{2} + \frac{m_{Li} v_{ic}^2}{2} = \frac{m_{Be} v^2}{2} \left(\frac{x}{x+1}\right)^2 + \frac{m_{Li} v^2}{2} \left(\frac{1}{1+x}\right)^2 = K \frac{x}{x+1}$$

$$K'_{\text{отн}} = \frac{m_{Be} (v'_{ic})^2}{2} + \frac{m_{Li} (v'_{ic})^2}{2}$$

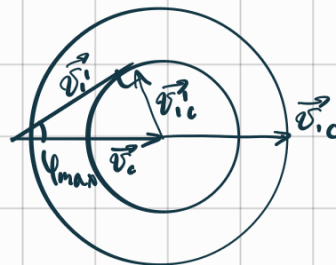
3C4

$$P_{\text{отн}} = m_{Be} v_{ic} + m_{Li} v_{ic} = 0$$

$$P'_{\text{отн}} = m_{Be} v'_{ic} + m_{Li} v'_{ic}$$

$$v_{ic} = -v'_{ic} \frac{m_{Be}}{m_{Li}} = -\frac{v'_{ic}}{x}$$

$$W = K \frac{v}{x+1} - \frac{m_{Be}}{2} \left(\frac{v \sin \psi_{\max}}{x+1}\right)^2 - \frac{m_{Li}}{2} \left(\frac{v \sin \psi_{\max}}{x(x+1)}\right)^2 = K \frac{x}{x+1} - K \frac{\sin^2 \psi_{\max}}{(x+1)^2} - K x \frac{\sin^2 \psi_{\max}}{x^2 (x+1)^2} =$$

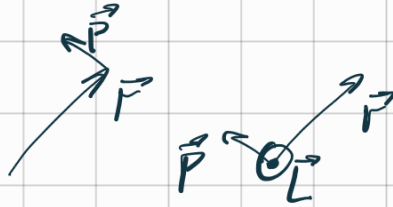


$$\sin \psi_{\max} = \frac{v'_{ic}}{v_c}$$

$$= K \frac{x^2(x+1) - \sin^2 \varphi_{\max}(x+1)}{x(x+1)^2} = K \frac{x^2 - \sin^2 \varphi_{\max}}{x(x+1)} \approx 6,5 \approx B$$

Побуждение

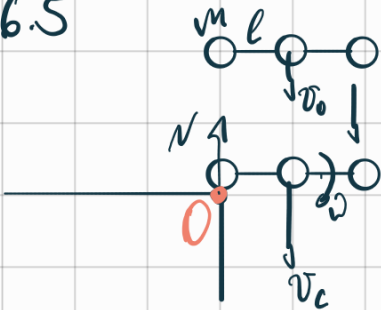
$$\vec{L} = [\vec{r} \times \vec{p}]$$



$$\frac{d\vec{L}}{dt} = [\vec{r} \times \vec{F}] = \vec{M}$$

$$\begin{aligned} \vec{M} = 0 &\Rightarrow \vec{L} = \text{const} \\ \vec{M}_x = 0 &\Rightarrow L_x = \text{const} \end{aligned}$$

6.5



ОТН. О $M_x = 0$

ЗСМН

$$\begin{aligned} m v_0 \cdot 0 + m v_0 \cdot l + m v_0 \cdot 2l &= \\ = m(v_c - \omega l) \cdot 0 + m v_c \cdot l + m(v_c + \omega l) \cdot 2l & \\ 3v_0 &= 3v_c + 2\omega l \end{aligned}$$

УПРЯГУ́Я ЧЛДП $\Rightarrow E = \text{const}$ ЗСЭ:

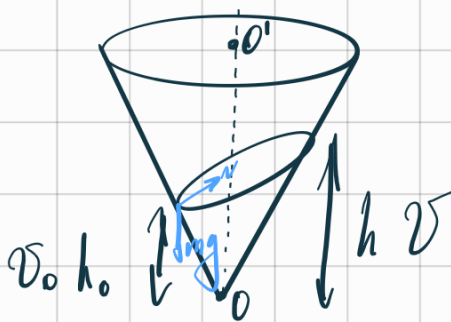
$$3m \frac{v_0^2}{2} = 3m \frac{v_c^2}{2} + 2m \frac{(\omega l)^2}{2}$$

$$\begin{cases} 3(v_0 - v_c) = 2\omega l \\ 3(v_0 - v_c)(v_0 + v_c) = 2(\omega l)^2 \end{cases}$$

$$\begin{cases} 3(v_0 - v_c) = 2\omega l \\ v_0 + v_c = \omega l \\ v_0 = \frac{5}{6}\omega l \\ v_c = \frac{\omega l}{6} = \frac{v_0}{5} \end{cases}$$

$$\omega = \frac{6}{5} \frac{v_0}{l}$$

6.8)



$$\frac{m v_0^2}{2} + m g h_0 = \frac{m v^2}{2} + m g h$$

$$\text{ОЦб } \text{ОО}' \quad M_{mg} = 0 \quad M_N = 0$$

$$L = \text{const}$$

$$m v_0 r_0 = m v r$$

$$\frac{v_0}{r} = \frac{h_0}{h}$$

$$v_0 h_0 = v h$$

$$v_0^2 + 2gh_0 = v^2 + 2gh$$

$$v^2 \left(\frac{h}{h_0} - 1 \right) = 2g(h - h_0)$$

$$v^2 = \frac{2g(h - h_0)}{\left(\frac{h - h_0}{h_0} \right)} = 2gh_0 \quad v = \sqrt{2gh_0}$$

$$v_0 = \sqrt{\frac{2gh_0}{h_0}} \cdot h = \sqrt{\frac{2g}{h_0}} \cdot h$$

$$\vec{F} = -G \frac{Mm}{r^3} \vec{r}$$

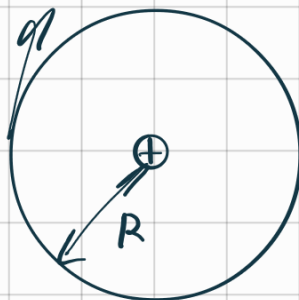
$$U = -G \frac{Mm}{r} + C$$

$$v_1 = \sqrt{G \frac{M}{R}}$$

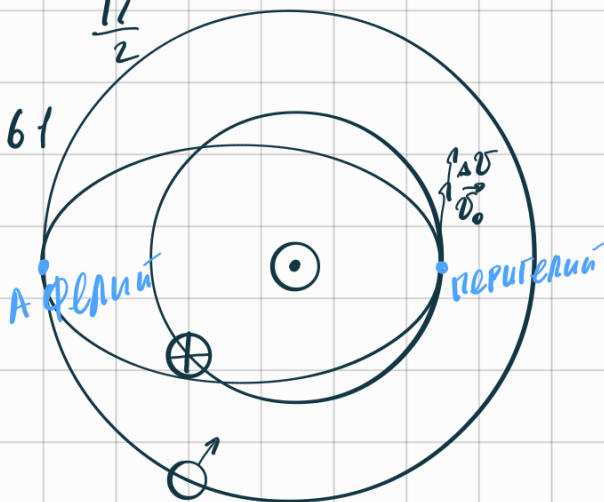
$$K = \frac{mv_1^2}{2} = G \frac{Mm}{2R}$$

$$E_u = \frac{-G Mm}{2R} = -K$$

$$E = -G \frac{Mm}{R}$$



7,61



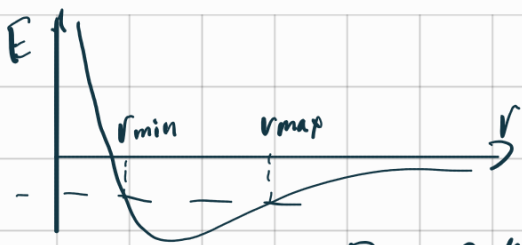
$$1. a. l. = 1,5 \cdot 10^{11} \text{ м}$$

$$v_0 \approx 30 \text{ км/с}$$

ЗСЭ

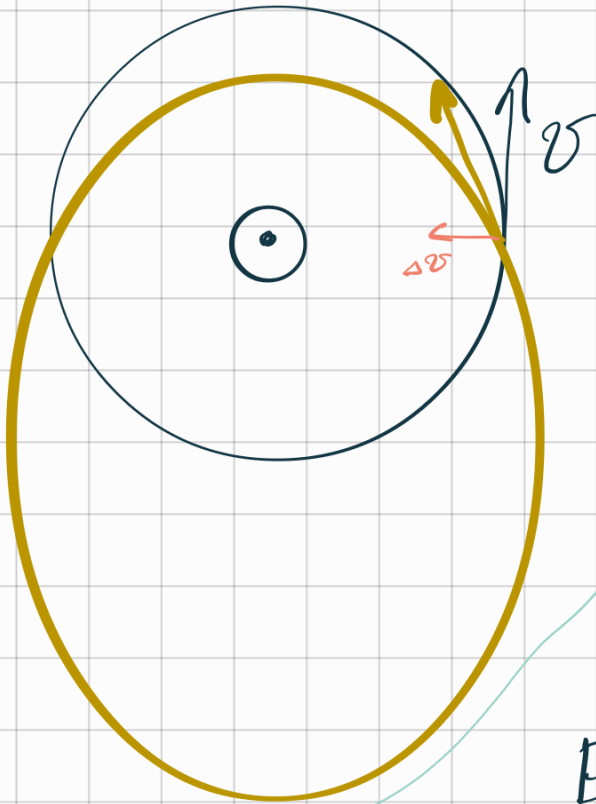
$$\frac{m(v_0 + 2v)^2}{2} - G \frac{Mm}{R_2} =$$

$$= \frac{mv^2}{2} - G \frac{Mm}{R_1}$$



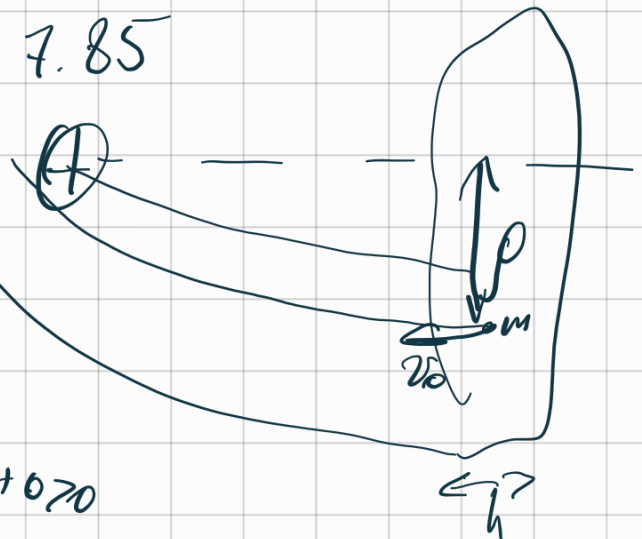
ЗЧМН ОТН 0
 $m(v_0 + \Delta v) R_3 = m v R_1$

$$E = -G \frac{Mm}{a} = -G \frac{Mm}{R_3 + R_1} \quad \tau \text{ рефз } 3 \text{ г.к.}$$



$$v_{II} = \sqrt{2} v_1 = \sqrt{\frac{2GM}{R}}$$

7.85



$$E = m \frac{v_0^2}{2} + 0.70$$

$$m \frac{v^2}{2} - G \frac{Mm}{R}$$

$$v = \sqrt{v_0^2 + \frac{2GM}{R}}$$

ЗЧМН:

$$m v R = m v_0 \rho \quad (\cdot \sin \alpha)$$

$$\rho = R \frac{v}{v_0} =$$

$$= R \sqrt{1 + \frac{2GM}{R v_0^2}} = R \sqrt{6}$$

$$N = n \cdot \pi \rho^2 \cdot h$$